

SIMATS ENGINEERING**Department of Computer Science and Engineering ASSESSMENT TOOL2 - CONCEPT MAPPING**

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO1	Assessment:	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	Total Marks:	25
CO1 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
Student Name:	JANESSA J	Reg.No.:192424017	
Department:	AIDS	Date:15/07/2026	

Instruction to Students

1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

Code of Conduct:

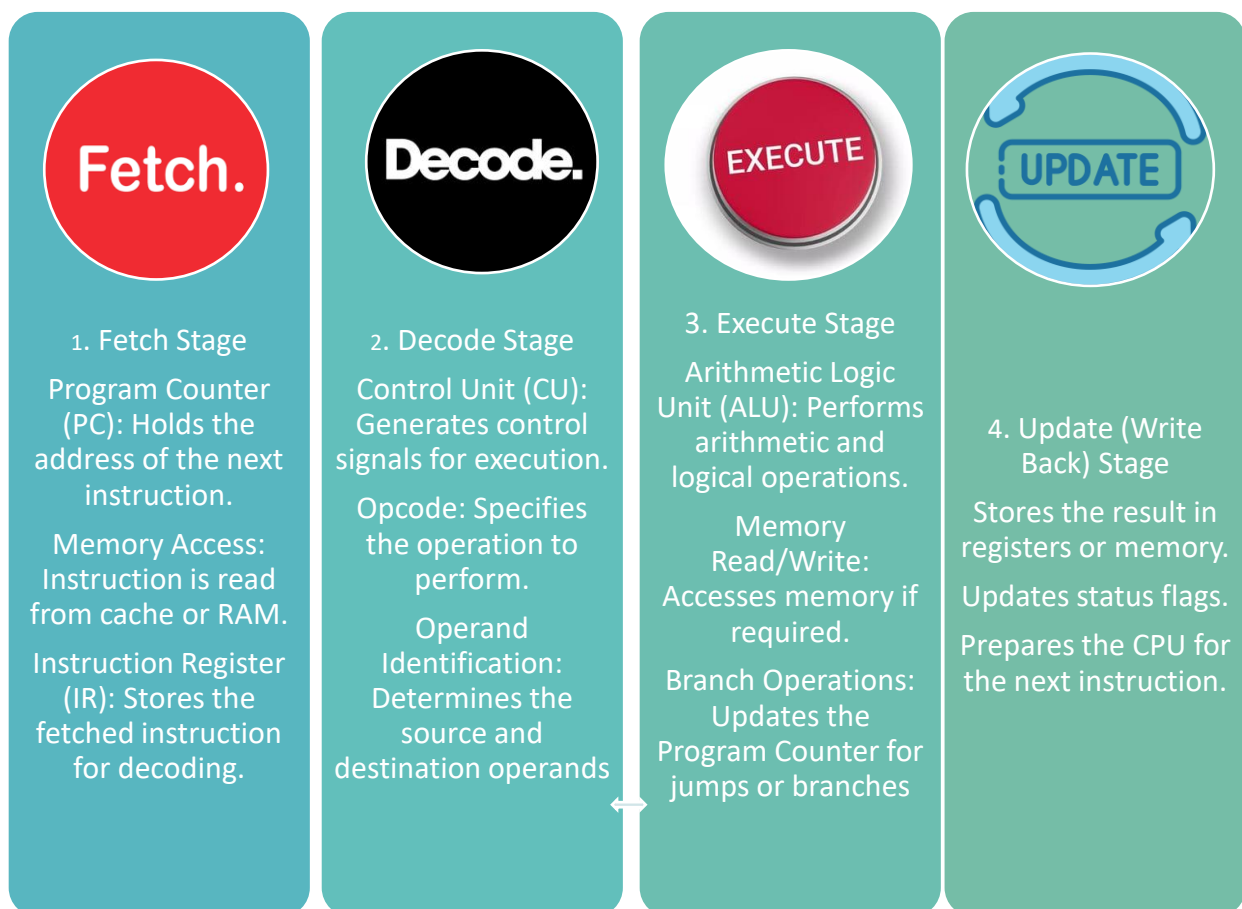
I JANESSA J (192424017)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

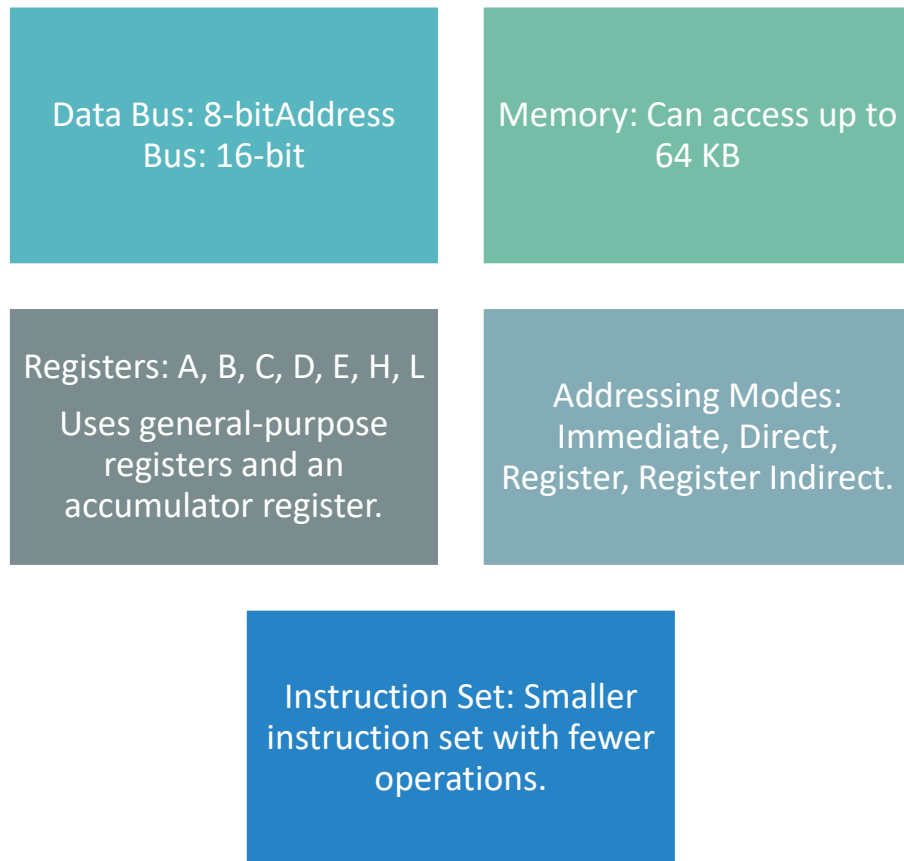
Janessa J

ANNEXURE A

1. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.



2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.



8086-MICROPROCESSOR

Data Bus: 16-bit

Address Bus: 20-bit

Memory: Can access up to 1 MB

Registers: AX, BX, CX, DX, SI, DI, BP, SP

Provides specialized registers for efficient processing.

Addressing Modes: Immediate, Direct, Indexed, Based, Based Indexed.

Instruction Set: Larger and more powerful instruction set.